Questions

Q1.

A bigger hole is made in the base of the same pool.

Compare the force on the ground directly below the bigger hole with the force on the ground directly below the original hole.

(1)

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(Total for question = 1 mark)

Q2.

Figure 15 shows a ball floating in seawater and the same ball floating in fresh water.

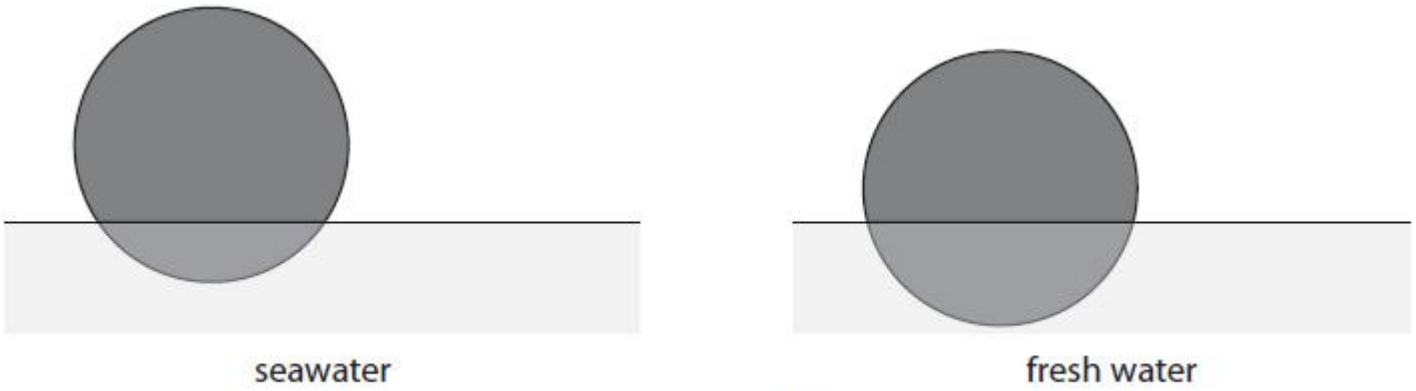

Figure 15

(i) Compare the upthrust on the ball in seawater with the upthrust on the same ball in fresh water.

(1)

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(ii) Explain why there is less of the ball below the surface of the seawater than below the surface of the fresh water.

(3)

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(Total for question = 4 marks)

Q3.

Figure 2 shows a tank for holding water.

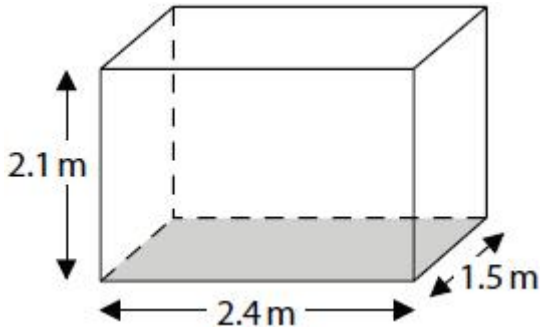

Figure 2

The tank has sides of 2.4 m, 2.1 m and 1.5 m.

The pressure at the bottom of the tank is 12 kPa.

(i) State the equation relating pressure, force and area.

(1)

(ii) Calculate the weight of water in the tank.

(4)

weight = ... N

(Total for question = 5 marks)

Q4.

Figure 14 shows an athlete using a fitness device.

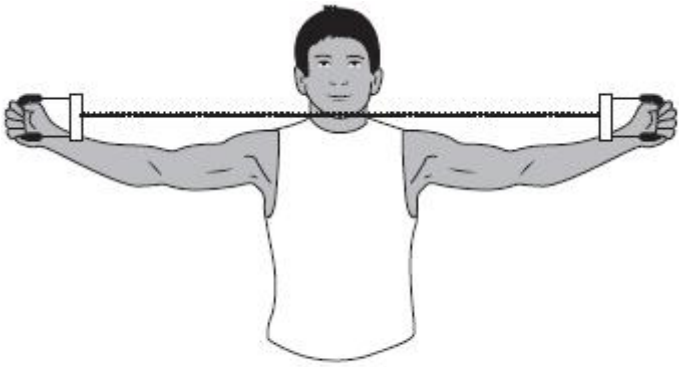

Figure 14

The athlete stretches the spring in the device by pulling the handles apart.

The spring constant of the spring is 140 N/m.

The athlete does 45 J of work to extend the spring.

The athlete takes 0.6 s to expand the spring.

(i) Calculate the useful power output of the athlete when stretching the spring.

(2)

useful power output of the athlete = ... W

(ii) Calculate the extension of the spring.

Use an equation selected from the list of equations at the end of this paper.

(3)

extension of the spring = ... m

(Total for question = 5 marks)

Q5.

A spring is extended.

A force of 0.50 N gives an extension of 13 mm.

Calculate the spring constant *k* in N/m.

(3)

spring constant *k* = ... N/m

(Total for question = 3 marks)

Q6.

Figure 16 shows a beaker containing 150 g of water.

Figure 16

The cross-sectional area of the bottom of the beaker is 3.3×10^{-3} m^{-2}.

Calculate the pressure at the bottom of the beaker due to the water.

Take gravitational field strength, g, to be 10 N/kg.

(2)

pressure = ... Pa

(Total for question = 2 marks)

Q7.

Figure 26 shows a glass U-tube containing water of density 1000 kg/m^3.

The water levels are the same on both sides of the U-tube.

Both ends of the U-tube are open to the atmosphere.

Atmospheric pressure is 101 000 N/m^2.

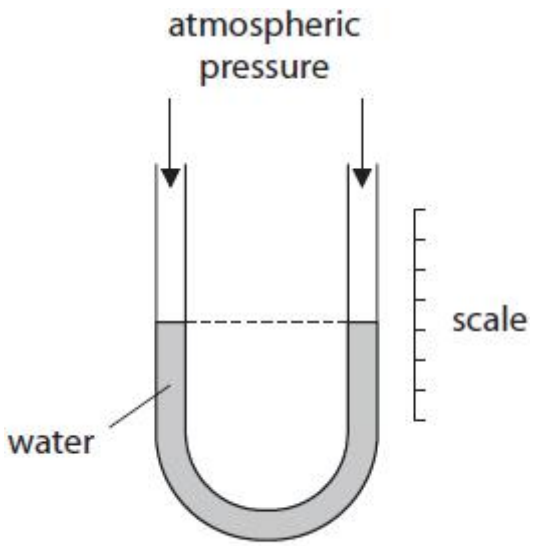

Figure 26

Figure 27 shows the U-tube with one side connected to a gas supply.

The difference in the levels of water, *h*, is 0.200 m.

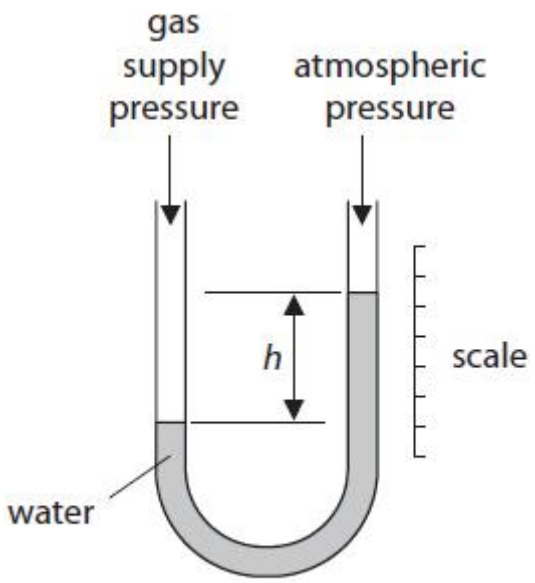

Figure 27

(i) Calculate the gas supply pressure.
Use an equation selected from the list of equations at the end of the paper.

(3)

pressure of gas supply = ... N/m^2

(ii) The measurement is repeated using a U-tube of larger cross-sectional area.
Explain why the value of *h* does not change.

(2)

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(Total for question = 5 marks)

Q8.

Figure 10 shows an ice skater standing on one skate.

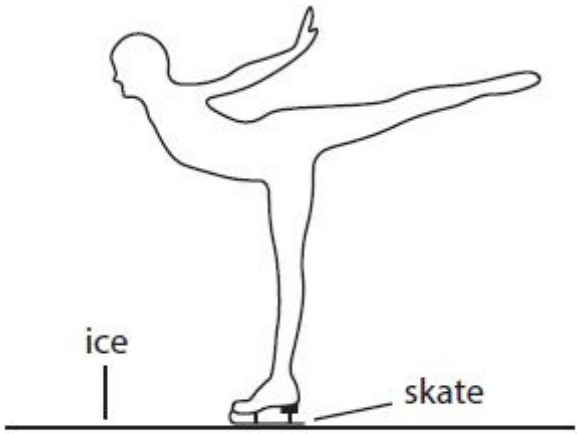

Figure 10

Calculate the force the skate exerts on the ice

pressure of skate on ice = 4.8×10^7 Pa

area of blade in contact with ice = 1.2×10^{-5} m^2

force = pressure × area

Give your answer to 2 significant figures.

(3)

force = ... N

(Total for question = 3 marks)

Q9.

Figure 1 shows a pool full of water.

(Source: © AnaKulagina/Shutterstock)

Figure 1

A small hole is made in the base of the large pool shown in Figure 1.

The area of the hole is 1.5 mm^2. The pressure at the base of the pool due to the water is 3500 Pa.

Calculate the force on the ground directly below the hole due to the water in the pool.

Use the equation

$$P = \frac{F}{A}$$

(2)

force = ... N

(Total for question = 2 marks)

Q10.

Figure 16 shows part of the inside of a pen.

The pen contains a spring that can be compressed.

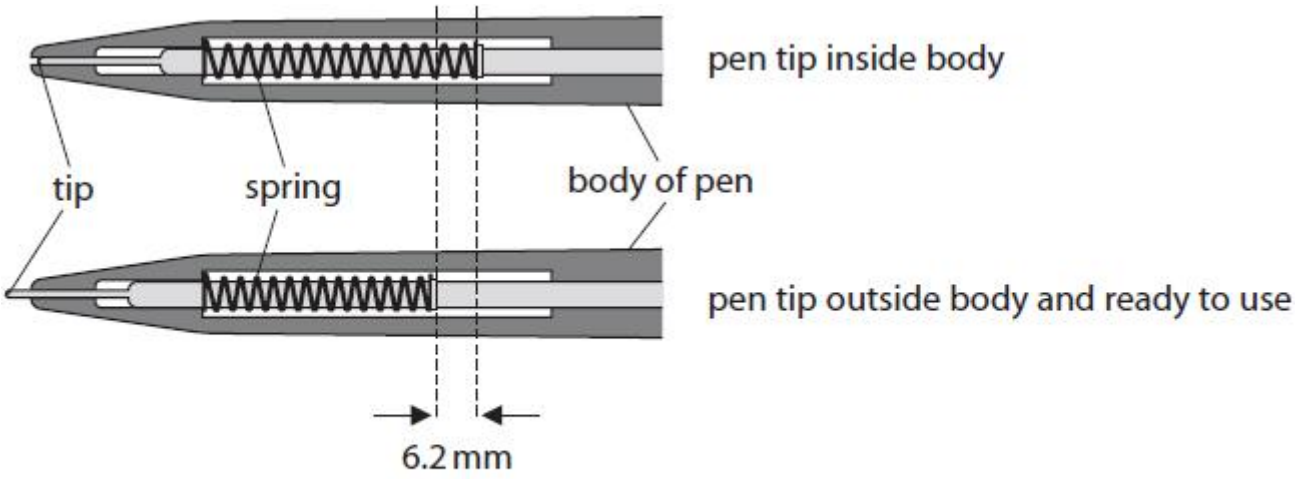

Figure 16

The spring constant of the spring is 260 N/m.

(i) Calculate the force needed to compress the spring by the amount shown in Figure 16.

Give your answer to an appropriate number of significant figures.

(3)

force = ... N

(ii) A student removes the spring from the pen and investigates the compression of the spring.

Figure 17 shows the equipment and the procedure that the student uses.

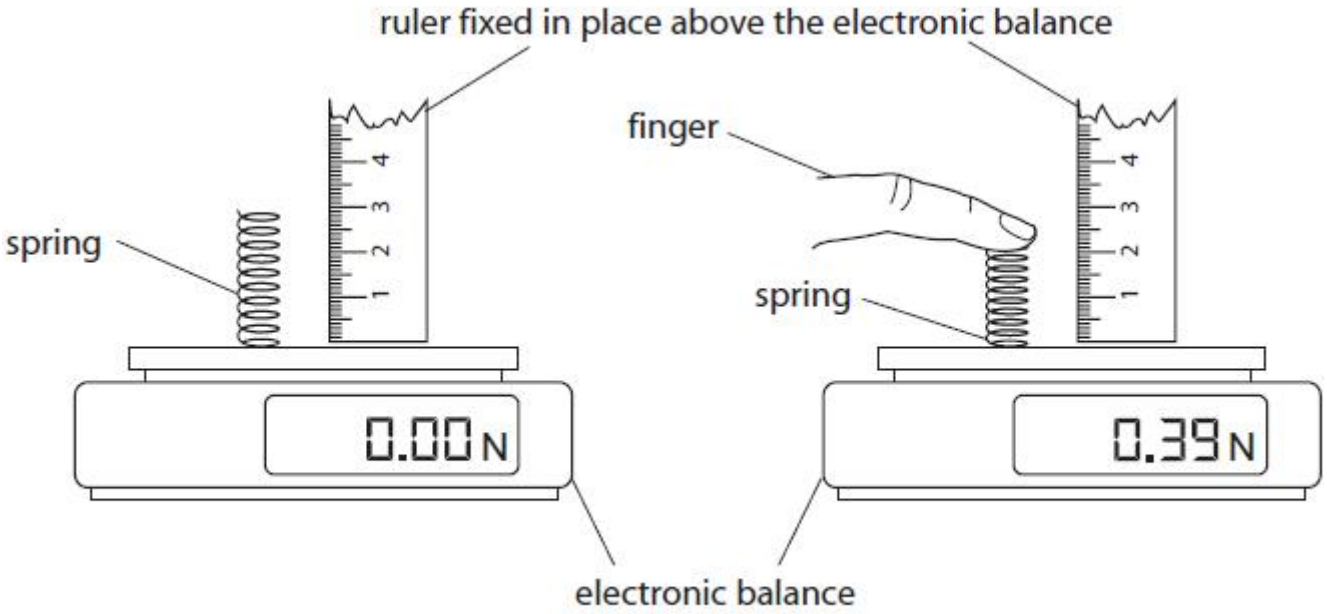

Figure 17

The student presses down on the spring to change its length.
The electronic balance measures the force applied to the spring.
Describe how the student can determine the change in length of the spring.
You may add to Figure 17 to help your answer.

(3)

(iii) The student finds it difficult to make an accurate measurement of the change in length of the spring using the equipment as shown.

Describe **one** way that the student could improve the procedure.

(2)

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.....

(Total for question = 8 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
	(force below the bigger hole is) larger (1)	accept force below original hole is smaller	**(1)** **AO1.1**

Q2.

Question number	Answer	Additional guidance	Mark
(i)	(upthrusts in each case) are equal (1)		**(1)** **AO3**

Question number	Answer	Additional guidance	Mark
(ii)	an explanation linking any **three** of weight of ball = weight of water displaced (1) seawater more dense than fresh water (1) smaller volume of seawater (needs to be) displaced (1) to produce same weight of water (displaced) (1)	accept reverse arguments accept saltwater for seawater upthrust = weight of water displaced accept less seawater displaced to produce same upthrust	**(3)** **AO1**

Q3.

Question number	Answer	Mark
(i)	pressure = force ÷ area	(1)

Question number	Answer	Additional guidance	Mark
(ii)	rearrangement (1) $(F =)\ P \times A$ calculation of area (1) $2.4 \times 1.5 = 3.6$ substitution (1) $(F =)\ 12\,000 \times 3.6$ answer (1) 43 200 (N)	award full marks for correct numerical answer without working maximum 3 marks if kPa not converted to Pa	(4)

Q4.

Question Number	Answer	Additional guidance	Mark
(i)	recall (1) $(P =)\ \frac{E}{t}$ substitution and evaluation (1) (P=) 75 (W)	P = work done ÷ time $P = \frac{45}{0.6}$ award full marks for the correct answer without working	(2)

Question Number	Answer	Additional guidance	Mark
(ii)	substitution into $E = \frac{1}{2} \times k \times x^2$ (1) $45 = \frac{1}{2} \times 140 \times x^2$ rearrangement (1) $(x =) \sqrt{\frac{2 \times 45}{140}}$ evaluation (1) 0.8(0) (m)	allow substitution and rearrangement in either order $x^2 = (\frac{E}{0.5k} =) \frac{2 \times 45}{140}$ $x^2 = 0.64(28571)$ accept values that round to 0.80 e.g. 0.80178 award full marks for the correct answer without working	(3)

Q5.

Question number	Answer	Additional guidance	Mark
	recall and substitution (1) $0.5 = k \times 13 (\times 10^{-3})$ rearrangement (1) $\frac{0.5}{13(\times 10^{-3})}$ evaluation (1) 38 (N/m)	 $k = \frac{F}{x}$ allow 38.5 (N/m) or 38.46 (N/m) or 39 (N/m) 0.04/0.038 (N/m) gains 2 marks 2958 (N/m) gains 1 mark (x^2 used in equation) award full marks for the correct answer without working	**(3)**

Q6.

Question number	Answer	Additional guidance	Mark
	recall and use of $P = \frac{F}{A}$ (1) evaluation (1) = 450 (Pa)	$P = \frac{0.15 \times 10}{3.3 (\times 10^{-3})}$ 454 (Pa)	**(2)**

Q7.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) (p) = 1000 x 10 x 0.200 evaluation of pressure difference (1) 2000 final evaluation (1) 103000 (Pa)	accept e.c.f for addition of atmospheric pressure seen for 1mark award 1 mark for selecting correct equation if no other mark awarded award full marks for correct answer without working.	(3)

Question number	Answer	Additional guidance	Mark
(ii)	an explanation linking use of $P = h \times \rho \times g$ (1) no area in the equation (1)	P /pressure, ρ /density (and g /gravitational field strength) are constant/the same Area does not affect result h /height of water is independent of area P, ρ, and g are all constant gains 2 marks	(2)

Q8.

Question number	Answer	Additional guidance	Mark
	substitution (1) (force =) 4.8 (x 10^7) x 1.2 (x 10^{-5}) evaluation (1) 576 (N) their evaluation rounded to 2sf (1) 580 (N)	award full marks for the correct answer (580) without working award 1 mark for 5.76 to any other power of ten award 2 marks for 5.8 to any other power of ten	**(3)** **AO2.1**

Q9.

Question number	Answer	Additional guidance	Mark
	substitution into $P = \frac{F}{A}$ (1) $3500 = \frac{F}{1.5 (\times 10^{-6})}$ rearrangement and evaluation (1) (F =) 0.0053 (N)	 5.3×10^{-3} (N) accept 0.0052(5) (N) or $5.2(5) \times 10^{-3}$ (N)	**(2)** **AO2.1**
	OR ALTERNATIVE METHOD rearrangement (1) F = P x A substitution and evaluation (1) (F =) 0.0053 (N)	 5.3×10^{-3} (N) accept 0.0052(5) (N) or $5.2(5) \times 10^{-3}$ (N) 0.0053 or 0.0052(5) to any other power of ten scores 1 mark award full marks for correct answer without working	

Q10.

Question number	Answer	Additional guidance	Mark
(i)	selection and substitution (1) (F=) 260 x 6.2 (x 10^{-3}) evaluation (1) (F=) 1.612 (N) or 1.61 (N) answer to 2 s.f. (1) 1.6 (N)	 award 1 mark only for answer of 1.61(2) to any other power of ten, e.g. 1612 (N) independent mark for any answer given to 2 significant figures allow 2 marks for answer of 1600 (N) with or without working 1.60 scores 2 marks award full marks for correct answer without working.	**(3)** **AO2**

Question number	Answer	Additional guidance	Mark
(ii)	a description including read position of top of spring against the ruler (1) read position of top of spring when pressed down (1) subtract the two readings (1)	May be seen drawn in figure 17 measure **length** at the start allow value from ruler e.g. 2.9 (cm) measure the **length** when pressed down allow value from ruler e.g. 2.0 (cm) subtract the two measurements allow find the difference for subtract allow calculated value from diagram e.g. 0.9 (cm) ignore repeat	**(3)** **AO1**

	OR substitution (1) 0.39 = 260 x change in length rearrangement (1) (change in length =) $\frac{0.39}{260}$ evaluation (1) 1.5 mm unit must be shown	 (0).0015m unit must be shown	

Question number	Answer	Additional guidance	Mark
(iii)	description to include change to enable accurate location of top of spring (1) for example: pointer, set square, thin sheet / another ruler (under finger) description of how the change is used (1)	may be seen drawn in Figure 17 move ruler closer to spring compress spring with weight rather than finger ignore photographs make measurements from where pointer / set square / thin sheet / other ruler touches the ruler reduce parallax error prevents fluctuations while measuring ignore repeats ignore unqualified references to accuracy or precision	**(2)** **AO3**